DATA

ROMAN REFERENCE CATALOG

Zhai et al 2021

GALAXIES (NO CMB MAPS)

RAW VERSION

Mass Cut $M_h > 10^{11}\, h^{-1} M\odot$

DUST FILTERED VERSION

Mass cut + $0.28685 F_{Ha} > 10^{-16}$ erg s^{-1} cm^{-2}

AGORA SIMULATION

Y. Omori 2022

HALOS **CMB MAPS**

METHODOLOGY

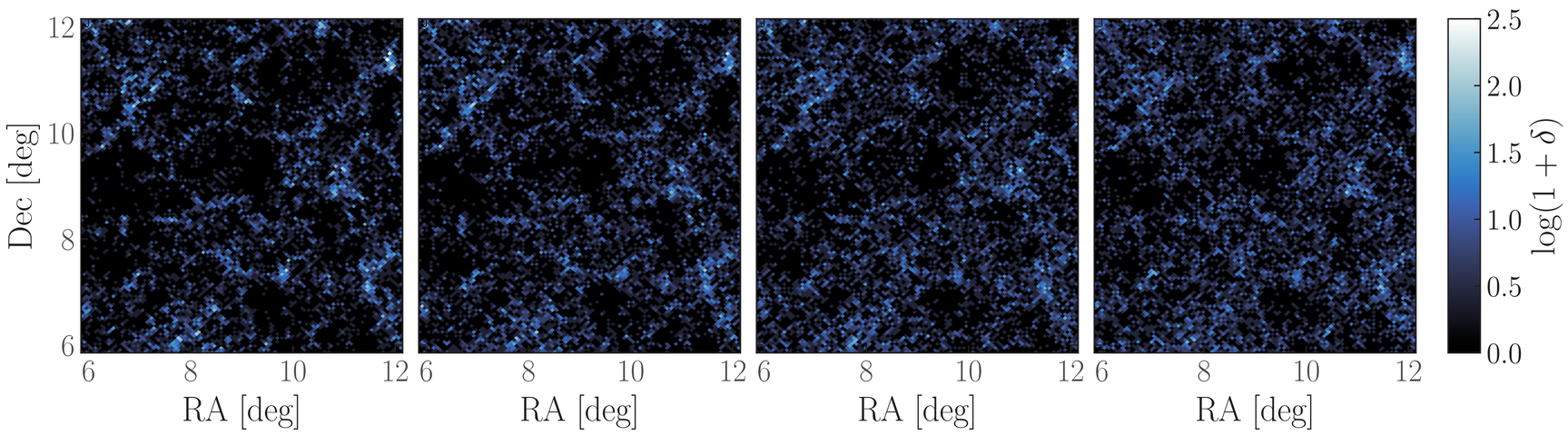
1. PIPELINE FOR GALAXY MOCKS GENERATION + VOID FINDING

INPUT: AGORA HALOS

PREPROCESSING

- Map baryonic-properties into halos
- Apply Roman mask
- Subsample halo catalog to the **Roman Raw** density using the SAM-augmented halo catalog. Four methods:
  - By stellar mass or randomly
  - All halos or ELG-host halos
- Complete the parameter space

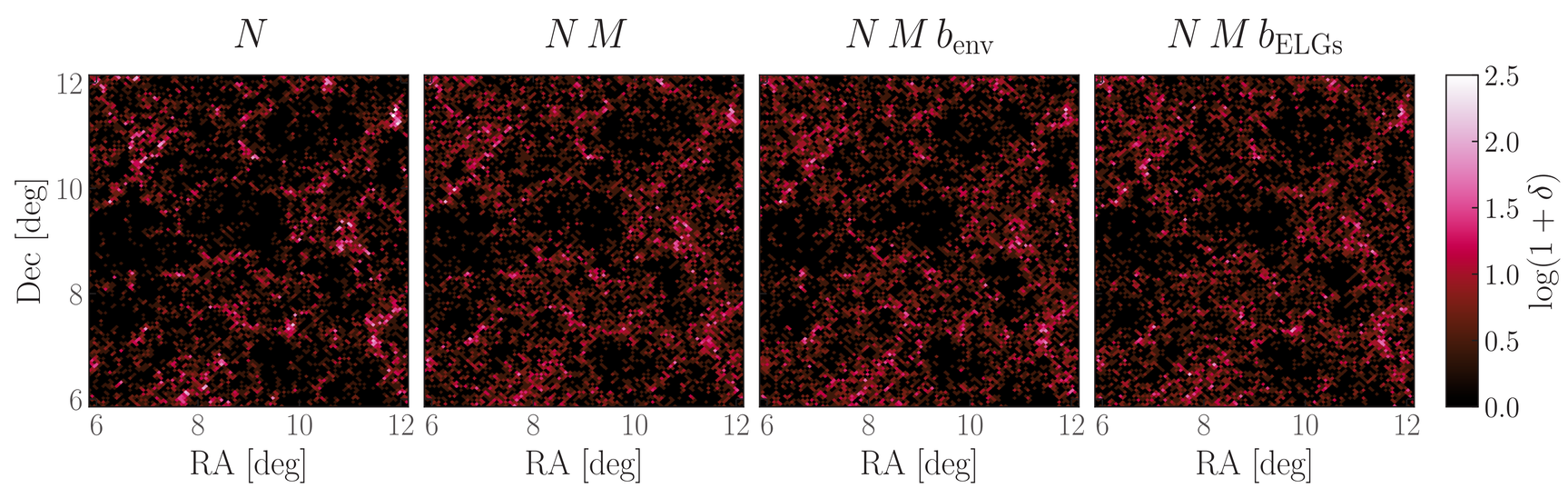
Different input files/biases



ANALOG MATCHING, AM

- Match each Roman galaxy **(dust filtered)** to its halo counter part based on the similarity of the parameter space and without repetition.
- 4 cases combining 4 parameters:
  N : number of objects (no AM)
  M : halo mass
  b_{env} : environment (δ_5)
  b_{ELGs} : galaxy type (M_*, sSFR)

Different AM options for each input file



VOID FINDING (2D AND 3D)

OUTPUT: GALAXY AND 2D/3D VOID CATALOGS